Problem Set 2

Applied Stats/Quant Methods 1

Due: October 22, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 22, 2026. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```
# I establish the hypotheses for independence.

print("H0: Class and police response towards traffic violations are independent.")
print("H1: Class and police response towards traffic violations are dependent.")

# First I calculate the expected frequencies

fe_upperc_not_stopped <- (27*21)/42
fe_upperc_bribe <- (27 * 13)/42
fe_upperc_warning <- (27 * 8)/42
fe_lowerc_not_stopped <- (15 * 21)/42
fe_lowerc_bribe <- (15 * 13)/42
fe_lowerc_warning <- (15 * 8)/42

# Here I enter the observed frequencies

fo_upperc_not_stopped <- 14
fo_upperc_bribe <- 6
fo_upperc_warning <- 7
fo_lowerc_not_stopped <- 7
fo_lowerc_bribe <- 7
fo_lowerc_warning <- 1

# Next I calculate each part of the chi squared function to be summed up, so the full equation will look neater

UC1 <- (fo_upperc_not_stopped - fe_upperc_not_stopped)^2/fe_upperc_not_stopped
UC2 <- (fo_upperc_bribe - fe_upperc_bribe)^2/fe_upperc_bribe
UC3 <- (fo_upperc_warning - fe_upperc_warning)^2/fe_upperc_warning
LC1 <- (fo_lowerc_not_stopped - fe_lowerc_not_stopped)^2/fe_lowerc_not_stopped
LC2 <- (fo_lowerc_bribe - fe_lowerc_bribe)^2/fe_lowerc_bribe
LC3 <- (fo_lowerc_warning - fe_lowerc_warning)^2/fe_lowerc_warning

# Finally, I sum up the above values to get the chi squared test statistic result

```

```r
drivers_chi_squared <- sum(UC1, UC2, UC3, LC1, LC2, LC3)
```

The above calculation provides a χ^2 test statistic result of 3.79.

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

```r
# I use an R command to calculate the p-value after deriving the degrees
    of freedom
df <- (2 -1) * (3 - 1)
pchisq(q = drivers_chi_squared, df = df , lower.tail = FALSE)

# I conclude as follows:

print("Because the p-value 0.15 is larger than the alpha level 0.1, we
    fail to reject the hypothesis that class is independent of police
    treatment in traffic.")
```

As indicated above, at a significance level of 0.1, we fail to reject the null hypothesis of independence.

[2]Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.3	-1.6	1.5
Lower class	-0.3	1.6	-1.5

```r
# First, I calculate the row and column proportions

UC_row_prop <- 27/42
LC_row_prop <- 15/42
Col_prop_not_stopped <- 21/42
Col_prop_bribe <- 13/42
Col_prop_warning <- 8/42

#Then I calculate the residuals for each cell as follows:

UC1_res <- (fo_upperc_not_stopped - fe_upperc_not_stopped) / sqrt(fe_upperc_not_stopped * (1 - UC_row_prop) * (1 - Col_prop_not_stopped))
UC2_res <- (fo_upperc_bribe - fe_upperc_bribe) / sqrt(fe_upperc_bribe * (1-UC_row_prop) * (1-Col_prop_bribe))
UC3_res <- (fo_upperc_warning - fe_upperc_warning) / sqrt(fe_upperc_warning * (1-UC_row_prop) * (1-Col_prop_warning))
LC1_res <- (fo_lowerc_not_stopped - fe_lowerc_not_stopped) / sqrt(fe_lowerc_not_stopped * (1-LC_row_prop) * (1-Col_prop_not_stopped))
LC2_res <- (fo_lowerc_bribe - fe_lowerc_bribe) / sqrt(fe_lowerc_bribe * (1-LC_row_prop) * (1-Col_prop_bribe))
LC3_res <- (fo_lowerc_warning - fe_lowerc_warning) / sqrt(fe_lowerc_warning * (1-LC_row_prop) * (1-Col_prop_warning))
```

(d) How might the standardized residuals help you interpret the results?

All of the standardized residuals are less than 2 and more than -2, indicating support for the null hypothesis of independence. In R, I noted my interpretation.

```r
print("The standardized residuals for the not-stopped-column are 0.3/-0.3, indicating support for the null hypothesis of independence. The standardized residuals for the bribe column are -1.6/1.6, and for the warning-column, 1.5/-1.5. These indicate at most a possible mild association. Overall, with standardized residuals falling below 2 and above -2, we fail to reject the null hypothesis.")
```

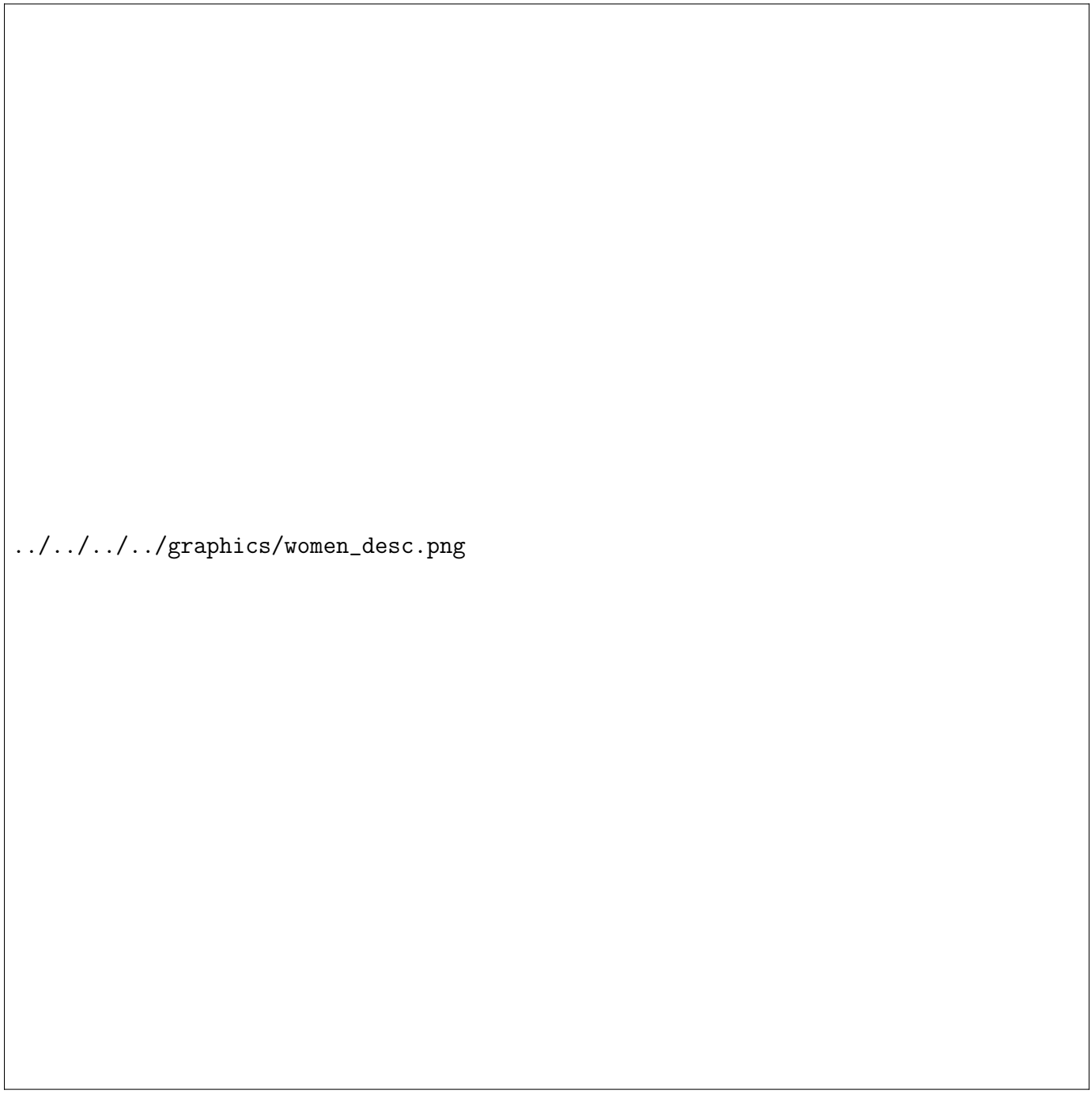
Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

[3]Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).



(a) State a null and alternative (two-tailed) hypothesis.

```
print("H0: The reservation policy's effect on number of new or repaired
    drinking water facilities equals zero.")
```

```
print("H1: The reservation policy's effect on the number of new or
    repaired drinking water facilities does not equal zero.")
```

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

```
plot(x=wpolicy$reserved, y=wpolicy$water)
cor.test(wpolicy$reserved, wpolicy$water)
model <- lm(water~reserved, data = wpolicy)
model
abline(model)
summary(model)

9.252/3.948
2*pt(q=2.34, df=320, lower.tail=FALSE)

print("As the p-value < 0.05, it indicates the slope coefficient is
    statistically significant, providing evidence against the null
    hypothesis that the effect of reservation policy on new or repaired
    drinking water facilities equals zero.")
```

Above, I have first plotted the variables and run a correlation test, the result of which (0.13) shows a weak positive correlation. Then I ran a regression in R with the reservation policy as the explanatory variable and new/repaired water facilities as the outcome variable. The regression provided me with the estimated coefficients to interpret below.

(c) Interpret the coefficient estimate for reservation policy.

The coefficient estimates for reservation policy are 14.738 for the y-intercept, and 9.252 for the slope. These values are interpreted as follows:

```
print("The estimated coefficients imply that, in the absence of
    reservation policy, 14.738 new or repaired water facilities are
    present. In the presence of reservation policy, an estimated 9.252
    additional new or repaired water facilities are available.")
```